Project Proposal - Code-Grounded Multi-Round AI Oral Assessment for STEM Coursework

Nov 7, 2026 · @Someone

Executive summary

We propose building and piloting a system where an AI examiner interviews students by voice about the code they submit, over three progressive rounds, so coursework counts as evidence of skill again even when AI can write the code.

Universities are moving STEM assessment back to exams, which protects grades but cuts hands-on practice. Meanwhile coursework no longer proves who did the thinking. Oral exams restore that link but never scaled. NYU Stern showed in two cohorts that voice AI can run and grade them for about $0.42 per student (source). Their stated next step is letting the examiner see student artifacts such as code (paper). That is the gap this project fills.

What we will build: a pipeline that reads a student's repository and git history, plans questions about their actual design decisions, thresholds and bugs, runs a 10-20 minute voice interview, and returns an evidence-linked grade for human audit.

What we ask for: a 16-week pilot on one programming course, with a build phase, one live assignment cycle, and a validity study against exam results. Marginal AI cost is expected to be a few dollars per student per round at NYU's reported rates; build time is the main cost (see Budget).

What success looks like: interview scores that track independent expert judgment, no unexplained subgroup gaps, student stress at or below NYU's second cohort, and a clear go/no-go for wider use.

Problem statement

STEM assessment is caught between two failures: exams that under-train practical skill, and coursework that no longer certifies it.

1. **Grade pressure pushes courses toward exams.** To control grade inflation and AI-assisted submissions, departments are shifting weight from coursework to closed-book exams.
2. **Exam-centric courses weaken hands-on skill.** Students spend less time designing, debugging and tuning real systems, which is the work employers expect.

3. **Coursework has lost its evidential value.** Students can paste assignment text into an LLM and submit polished code. Detectors are unreliable, and institutions have largely stopped relying on them (NYU paper). One course reported that students used auto-graders by trial and error to reach high marks without engaging (Conversational Exam).
4. **Oral assessment is valid but does not scale.** A 25-minute oral reviewed by two graders took roughly 30 staff hours for 36 students in the NYU course, and becomes untenable at 100 (paper).

The core question: can we keep project-based coursework and still verify that each student understands what they built, at a cost a department can sustain?

Proposed solution

Keep the project, and make the student defend it: an AI examiner reads each student's code and history, then interviews them about the decisions it contains.

The approach rests on three ideas.

1. **Design coursework so decisions emerge.** A well-built assignment forces choices that have no single right answer: thresholds, error handling, data structures, trade-offs between approaches. Real bugs surface during implementation. These become the raw material for questions.
2. **Ground the interview in the student's own artifact.** The examiner receives the repository, commit history and test results, and asks about this student's path (for example, why a retry limit changed between commits). Generic questions are avoided because the plan is built per student.
3. **Assess in rounds, as real engineering works.** The student presents an MVP, extends it, then discusses scale and technology. Later rounds check earlier claims against what was actually built, which makes rehearsed or outsourced work hard to sustain.

The AI conducts and drafts grades for each interview. Humans set the rubric, audit flagged and sampled cases, and own every consequential mark. The format follows the evidence that validity should come from the assessment's design rather than from policing behaviour (Conversational Exam).

Related work

No existing project combines a voice examiner, submitted code and history as input, and progressive rounds; each covers part of it.

Project	What it does	Uses student code	Multiple rounds
NYU Stern Viva	ElevenLabs voice agent runs a project defence and case discussion; three LLMs grade the transcript	Project summaries only	No
Conversational Exam	Oral exam where students code live while explaining their reasoning	Live coding, not submissions	No
UNSW prototype	Ingests programming submissions, asks personalised questions, gives provisional rubric evaluation	Yes	Not described
TAMIGO	LLM generates viva questions and feedback to help teaching assistants (GPT-3.5)	Yes	No
HackerRank Chakra	AI interviewer for hiring: voice/video interview in a live coding environment, evidence-backed scorecards	Live coding	No
This proposal	Voice examiner grounded in repository and git history, three rounds that build on each other	Yes, with history	Yes

Two lessons carry over directly. NYU's deployment shows what breaks in practice (stacked questions, hostile voice, non-random selection) and which fixes work. The hiring-side results suggest AI-led voice interviews can match human interviewers; the NYU authors cite a field experiment reporting higher offer and retention rates (paper), though we have not verified that study ourselves.

The Conversational Exam authors also flag open problems we share: disability accommodation, cultural norms around oral assessment, and the upfront cost of preparation (summary).

Assessment design

Each assignment is assessed in three short interviews, and each round reads what the student built since the last one.

Round	Input to the examiner	Focus of questions	Length	Proposed weight
1. MVP review	Repository, tests, commit history at MVP	Design decisions, thresholds, bugs met, alternatives rejected; student states plan for next iteration	10-15 min	30%
2. Extension	Diff since round 1, round 1 transcript and stated plan	What changed and why, how the plan held up, refactoring, one small "what breaks if..." modification	10-15 min	35%
3. Scale and technology	Final repository	Bottlenecks at 10x and 100x load, technology trade-offs, what they would redesign	10-15 min	35%

Weights are a starting point to tune in the pilot. Round 3 is mostly verbal, so it is graded only on ideas the course actually taught.

Coursework design principles

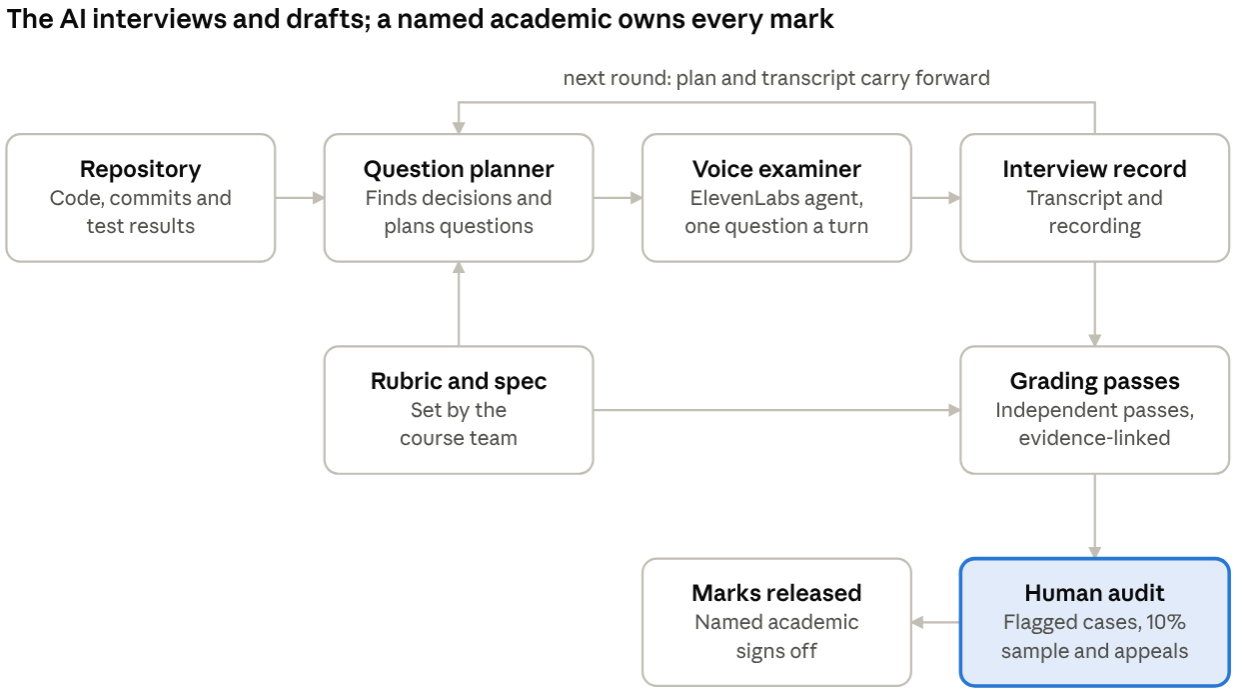
- Specify outcomes, not an implementation, so several valid solutions exist.
- Include at least three decision points with real trade-offs (a threshold, a data structure, a failure-handling policy).
- Use data or requirements that make naive solutions fail in instructive ways, so bugs emerge naturally.
- Require commits throughout, so process evidence exists.
- Publish the interview format, rubric and a practice run before the first graded round.

Scaling by level. First-year courses use round 1 and a light round 2, with round 3 replaced by a reflective "what would you improve" discussion. Final-year and masters courses use all three rounds.

Mark scheme. The code is credited in proportion to the student's demonstrated understanding, so a strong submission the student cannot explain scores low. Marks for each round are independent, so one weak round does not lock in a penalty, and honest self-correction scores well.

System architecture

The pipeline turns a repository and a rubric into an interview, a draft grade and a human-reviewed mark.



system architecture · 8 components, 1 human gate

The planner reads the repository and rubric. The examiner runs one round. Grading passes score the transcript. A human audits before marks are released, and the next round starts from this round's plan and transcript.

Component	Built with	Notes
Question planner	OpenAI model plus static analysis of the repository	Extracts decision points, thresholds and error handling; any random choice is drawn in code
Voice examiner	ElevenLabs agent with workflow phases and per-student dynamic variables	One-question-per-turn check, long silence tolerance, calm voice, live transcript shown to the student
Interview record	ElevenLabs post-call webhook plus our own storage	Transcript and recording with retention limits
Grading passes	OpenAI models	Independent passes, a deliberation pass, then a synthesis with quoted evidence

Component	Built with	Notes
Audit and release tool	Web app	Queue built from the audit rule; evidence view; appeals
Student portal	Web app	Upload code, practice runs, transcript, feedback report

ElevenLabs provides workflows, dynamic variables and post-call webhooks, which is how NYU built its examiner. OpenAI's Realtime API is the fallback voice layer: gpt-realtime-2.1 was released on 6 July 2026. The NYU authors note that multi-agent routing and per-session variables would then need rebuilding in our own code (paper).

Grading and rubric

Every score must point to evidence in the transcript, and a human audits the cases where the graders disagree or the stakes are high.

Rubric. Five dimensions scored 0-4, following the structure NYU used (paper), adapted for code:

Dimension	What a strong answer shows
Decision rationale	Explains why a threshold, structure or approach was chosen and what was traded off
Code understanding	Accurately describes what specific parts of their own code do and why
Debugging and failure handling	Describes real bugs met, how they were found and fixed, and what remains fragile
Consistency across rounds	Plans, claims and later changes match the history (rounds 2 and 3)
Communication under follow-up	Answers the question asked, adapts when challenged, admits uncertainty honestly

Grading process.

1. Several independently prompted grading passes score the transcript against the rubric, each quoting transcript evidence.
2. A deliberation pass lets each grader revise after reading the others' evidence. NYU found this raised agreement sharply, with perfect three-way agreement rising from 29% to 74% in its first cohort.

3. A final synthesis pass produces the score and a student-facing feedback report.
4. If the stacked-question protocol triggers (the examiner asked several questions at once), graders score only the parts the student answered.

Human audit rule.

- Review every exam where graders differ by more than one point on the total, or where one grader is far from the others.
- Review a random 10% sample of the rest each round, plus every appeal and every suspected integrity case.
- No mark is released without a named academic accountable for it.

NYU's rule flagged about 3% of exams (paper); we will measure our own rate.

Single-vendor caveat. NYU graded with models from three different providers, and chose them partly because they disagreed in useful ways. With only OpenAI access, diversity must come from different models, prompts and reasoning settings, which is weaker. We will compare against human marks on a calibration set and add a second provider if the gap is material.

Calibration. Before the pilot, two staff independently mark about 20 recorded practice interviews to tune the rubric and set a baseline for agreement.

Fairness, accessibility and integrity

A live oral can disadvantage students who understand the material but speak slowly or anxiously, so accommodations and a fairness analysis are built in, not added later.

What the NYU deployment found (paper):

- 83% of first-cohort students found the AI exam more stressful than written exams, falling to 63% after fixes. Stress was tied to stacked questions, short silence timeouts and a harsh cloned voice.
- Students with anxiety accommodations sometimes found the AI examiner less stressful than a senior professor.
- International students reported the barrier was articulating thoughts under pressure in English, not being understood by the system.
- The authors performed no subgroup fairness analysis and claim reliability, not validated fairness.

Our commitments

- Configurable time limits and pacing, a choice of voice and speaking rate, and a silence tolerance of at least 30 seconds.

- A live transcript so students can catch mis-hearing, and a text-based alternative for those who cannot use voice.
- An ungraded practice run and published rubric before any graded round.
- A calm, neutral voice. We will not clone a staff member's voice.
- Spot-checking of speech recognition accuracy on accented speech against gold transcripts.
- Subgroup analysis of scores and stress (accent, first language, declared accommodations) in the pilot, with small groups handled carefully and reported with their uncertainty.

Integrity. Dynamic, per-student questions defeat leaked question banks and rehearsed scripts. Gaps that remain: live AI coaching through an earpiece cannot be fully prevented, and an off-camera helper is only deterred (paper). We will use recorded sessions and the consistency check across rounds as deterrents, and state plainly that no mitigation is absolute.

Authenticity risk. Students may rehearse against the published format. That is acceptable if it forces real understanding, which is why questions vary by student and round.

Regulation and data governance

The system evaluates learning outcomes, so we design it as a high-risk system from day one: human decision-makers, disclosure to students, and a data protection impact assessment. This section is a map of what to check with the university's data protection and legal teams, not legal advice.

UK

- Under UK GDPR, students can object to purely automated decisions and ask for human review, and a Data Protection Impact Assessment may be required (policy review). Our design keeps a named academic accountable for every mark.
- University guidance on AI in marking typically requires a subject expert to review and approve AI output before release, and transparency with students (Exeter guidance).
- Human review only protects students if reviewers have time and authority to overturn results (IT Brief UK). A UK trial found oversight of AI marking did not automatically reduce workload (report).

EU (if EU students are served)

- AI used to evaluate learning outcomes is classed high-risk under Annex III of the EU AI Act. Obligations were deferred to 2 December 2027, but chatbot-disclosure duties applied from 2 August 2026 (summary; law firm note).

Data flows and safeguards

- Voice audio and transcripts go to the voice provider, and transcript text goes to the grading provider. NYU relied on provider privacy policies and, where available, enterprise or zero-retention agreements, and did not audit them independently (paper).
- ElevenLabs lists a per-agent zero-retention mode (changelog); we will confirm current terms and region options before any student data is used.
- Collect the minimum: transcripts and scores, with pseudonymous student IDs. Set retention limits, and delete audio after the appeal window.
- Disclose AI use, recording and data recipients in advance, with consent routes and an equivalent alternative for students who object.
- Record the DPIA, vendor agreements and the human-review procedure before the pilot starts.

Pilot plan and evaluation

The pilot runs on one programming course for 16 weeks, starts low-stakes, and is judged mainly on whether interview scores match independent expert judgment.

Design

- **Cohort:** one course with roughly 40-100 students. Open questions for the course lead: which course, how many students, and whether a parallel section or prior cohort can serve as comparison.
- **Stakes:** the first graded cycle counts for a small share of the coursework mark. Weight rises only if the validity checks pass.
- **Humans in the loop:** all marks are reviewed under the audit rule above, and students can request human review of any result.
- **Validity study:** a random 20% of interviews are marked blind by two staff from the recording and compared with the AI-drafted marks. NYU names this as the missing step to show validity (paper).

Measures

Measure	Target	Baseline
Agreement between AI and blind human marks	Krippendorff's alpha of 0.80 or higher on total score	NYU council reached 0.83 and 0.95 among AI graders, not against humans

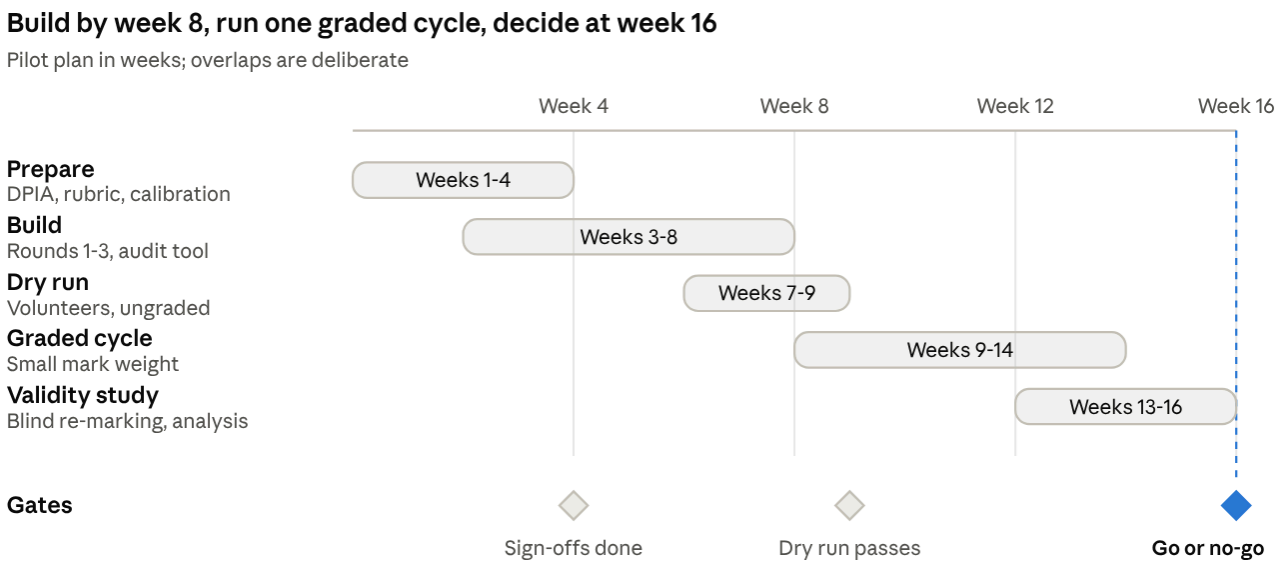
Measure	Target	Baseline
Interview score vs independent exam performance	Moderate positive correlation, reported with intervals	None published for code-grounded interviews
Stacked-question rate	Under 10% of examiner turns	NYU: 31% falling to 11%
Students finding it more stressful than written work	Under 60%	NYU: 83% falling to 63%
Students finding questions clear	80% or higher	NYU: 33% rising to 78%
Exams flagged for human audit	Measured, expected a few percent	NYU: about 3%
Subgroup score gaps (accent, language, accommodations)	No unexplained gaps	Not measured at NYU
Failed or interrupted sessions	Under 3%	Not reported
Staff time per student per round	Under 10 minutes including audits	NYU comparator: about 30 staff hours for 36 students by hand

The NYU baselines come from the paper; targets are our proposals and should be agreed with the course lead.

Decision at week 16. Go to wider use only if agreement and fairness targets are met and student stress is no worse than NYU's second cohort. Otherwise, keep the system as a formative practice tool and fix the specific failures.

Timeline and milestones

The pilot runs 16 weeks in five overlapping phases, with a go or no-go decision at the end.



pilot timeline · 5 phases, 3 gates, 16 weeks

The build overlaps preparation so a dry run with volunteers can start by week 7. The graded cycle starts only if the dry run passes: low rate of stacked questions, working accessibility options, and a signed-off data protection assessment. Week numbers are a plan to adjust once a start date and course calendar are fixed.

Budget and cost model

AI usage is small, at roughly US$1,500-3,000 for a 100-student pilot, so the real cost is staff time to build, audit and validate.

NYU's planning figure above its subscription was about US$0.11 per student-minute, or about US$3.40 for a 30-minute graded exam (paper). At 15 minutes per round, three rounds cost about US$5 per student per assignment. All figures below are approximate planning estimates, to be replaced with the course lead's numbers and current vendor pricing.

Item	Estimate	Basis
Voice and grading usage, 100 students, 2 assignment cycles (dry run and graded), 3 rounds each, plus 1 practice run per cycle	US$1,500-3,000 including a 100% safety margin	NYU planning rate of US$0.11 per student-minute; OpenAI grading costs to be measured

Item	Estimate	Basis
Voice platform plan	One monthly plan for 4 months; NYU's US$99 per month plan covered voice minutes in its cohorts	NYU paper; confirm current ElevenLabs plans and limits
Engineering	About 1 developer for 10-12 weeks	Our estimate: repository ingestion, question planner, examiner workflow, grading and audit tools, student portal
Assignment and rubric design	About 3-5 days per assignment	Our estimate; NYU needed 4-8 hours for a simpler exam and rubric
Human audit	About 15-20 hours over the pilot	About 15% of ~600 interviews at ~10 minutes each
Blind validity study	About 60 hours	20% of interviews, two markers, ~15 minutes each
Data protection and legal review	About 1-2 weeks of specialist time	Our estimate: DPIA, vendor terms, student notices
Accessibility testing	5-10 students with accommodations, plus staff time	Our estimate

Staff costs depend on local pay rates and are not priced here. Voice minutes are the variable cost to watch: usage grows with students, rounds and practice runs, and re-takes after technical failures also consume minutes.

Risks and mitigations

The highest risks are wrong or unfair marks and student distress; both are addressed by audit, transparency and a staged rollout. Likelihood and impact ratings are our judgement.

Risk	Likelihood	Impact	Mitigation
Examiner asks several questions at once or drifts off topic	High	Medium	Single-purpose phases and a code-level check that allows one question per turn; NYU cut stacking from 31% to 11% this way (paper)
AI misreads the code and asks wrong or unfair questions	Medium	High	Questions generated from a per-student plan linked to specific files and commits; rubric and spec supplied as context; test on a calibration set; human review of sampled question plans
Grading is unreliable or biased	Medium	High	Multiple grading passes, evidence-linked scores, human audit rule, blind validity study before weight increases
Students find it too stressful or unfair	High	Medium	Practice run, published rubric, calm voice, long silence tolerance, text alternative, short interviews
Accent or disability disadvantages some students	Medium	High	Live transcript, accommodations, accuracy spot-checks on accented speech, subgroup analysis in the pilot
AI-written code or live AI coaching goes undetected	Medium	Medium	Process evidence from commit history, consistency checks across rounds, recorded sessions; residual risk stated openly
Data protection non-compliance	Medium	High	DPIA and vendor review before any student data is used; minimal data; retention limits; disclosure and alternatives
Human review becomes a rubber stamp	Medium	High	Named accountable academic, authority to overturn, audit hours budgeted in the plan

Risk	Likelihood	Impact	Mitigation
Vendor outage, price change or policy change	Medium	Medium	Keep prompts and orchestration portable; NYU notes the core voice capabilities are common across platforms (paper); fallback to rescheduled or text-based interview
Randomisation or selection left to the language model is biased	High	Medium	Draw random choices in code, not in prompts; NYU's agent picked one case in 86% of exams until this was fixed (paper)

Success criteria, next steps and sources

The pilot succeeds if the measures in the evaluation table are met and a go/no-go can be made at week 16 on evidence rather than enthusiasm.

Next steps

- [ ] Choose the pilot course and confirm cohort size and comparison group
- [ ] Agree rubric, round weights and audit rule with the course lead
- [ ] Start the DPIA and vendor review before any student data is used
- [ ] Build round 1 on one assignment and run it with volunteers, ungraded
- [ ] Double-mark about 20 practice interviews to calibrate the rubric
- [ ] Confirm current ElevenLabs plan limits and OpenAI model choices and pricing

Open questions

- Which course and institution, and which local assessment and AI policies apply?
- Is OpenAI the only grading provider available, or can a second be added?
- Is a text-only fallback acceptable for students who opt out of voice?

Sources. The NYU paper was read in full; the other pages were read as search excerpts and should be re-checked before citing formally. Figures marked as estimates are ours.

- Scalable and Personalized Oral Assessments Using Voice AI (NYU, arXiv)
- NYU professor oral exams for 42 cents per student (The Decoder)
- The Conversational Exam (arXiv)
- UNSW automated oral assessments prototype
- TAMIGO (arXiv)
- HackerRank's AI interviewer Chakra (TechCrunch)

- EU AI Act education deadline deferred
- EU Digital Omnibus on AI enters into force (K&L Gates)
- Policy review of AI-based contribution evaluation, incl. UK GDPR (arXiv)
- Exeter guidance on AI in marking and feedback
- When AI grades student work, who checks the feedback? (IT Brief UK)
- AI marking oversight trial in UK higher education
- OpenAI gpt-realtime-2.1 release (DataNorth)
- ElevenLabs agents documentation